

CT 303: Digital Communication

DAU (Formerly DA-IICT), Autumn Semester 2025-2026

Instructors (sections A & B in order): Dr. Manish Kumar & Dr. Abhishek Jindal

Course Credits (L-T-P-C): 3-0-2-4

Course Level (UG/PG): Under-Graduate (UG) - For B.Tech (ICT, ICT-CS) program, 5th semester

Lecture Timings: TBD

Laboratory Timings: TBD

Prerequisites: CT 203: Signals and Systems, CT 111: Introduction to Communication Systems, SC224: Probability & Statistics and SC 217: Electromagnetic Theory

Course Contents:

- **Module 1: Introduction to digital communication systems and digitization of analog signals**
 - Analog versus digital communications, overview of digital communication based system, review of sampling theorem, ideal and practical sampling, aliasing, analog signal reconstruction from discrete-time samples, pulse code modulation (PCM): uniform and non-uniform quantization and companding, and Differential PCM.
- **Module 2: Digital modulation**
 - Signal space concepts: representation of signals as vectors and Gram-Schmidt orthonormalization. Signal representation and constellations: amplitude shift keying (ASK), phase shift keying (PSK), rectangular and non-rectangular quadrature amplitude modulation (QAM) and frequency shift keying (FSK). Design for bandlimited channels: power-bandwidth tradeoff and Nyquist criterion for ISI avoidance.
- **Module 3: Digital demodulation and detection**
 - Optimal demodulation in additive white Gaussian noise (AWGN): maximum-likelihood (ML) decision rule and minimum probability error (MEP) decision rule. Realization of optimal receiver using matched filters. Performance analysis of ML reception and link-budget analysis.
- **Module 4: Elements of Information theory**
 - Notion of channel capacity, capacity of discrete-time AWGN channel: sphere packing interpretation, capacity of band-limited AWGN channel, power-bandwidth tradeoff in bandlimited AWGN channel, and design implications of Shannon limits.

Assignments & Laboratory: The course has both the components. Analytical problems will be regularly given to be solved. In laboratory, the concepts learned in the lectures need to be validated through hardware and software based experiments. For the software part, performance analysis will be the primary focus.

Course References:

1. *Communication Systems*, Simon Haykin, publisher: Wiley, Fourth Edition, 2001.
2. *Digital and Analog Communication Systems*, Leon W. Couch, publisher: Pearson, Eighth Edition, 2014.
3. *Modern Digital and Analog Communication Systems*, B. P. Lathi, Zhi Ding & Hari M. Gupta, publisher: Oxford University Press, 4th edition, 2017.

4. *Digital Communications*, J G Proakis & M A Salehi, publisher: McGraw Hill, Fifth Edition, 2008.
5. *Introduction to Communication systems*, U. Madhow, publisher: Cambridge university Press, 2014.
6. *Digital Communications: Fundamentals and Applications*, 2nd edition, B. Sklar and P. B. Ray, LPE, publisher: Pearson Education, 2009.
7. *Communication Systems Engineering*, 2nd edition, J. G. Proakis and M. Salehi, LPE, publisher: Pearson Education

Evaluation Policy:

First In-Semester Examination (20 August to 23 August)	23%
Second In-Semester Examination (03 October to 07 October)	24%
End-Semester Examination (21 November to 29 November)	28%
Laboratory Work (Report Submission + Lab attendance + Viva)	10%
Lab Test	15%

Attendance Policy: Attendance will be marked in each lecture. At the end of the semester, if a student has an overall attendance of less than 70% and he/she has scored a total marks of Y out of 100, then the total marks will be recalculated as $Y - \min(5, 70 - \text{your attendance in } \%)$. Note that the 30% of absence includes all medical and co-curricular-activities related leaves.

Course Outcomes: After completing the course, students will be able to:

- Understand transmission and reception of discrete data.
- Study effect of noise on the communication systems.
- Quantify information and decide on the resource required to transmit and receive data.
- Simulate communication systems and do a performance study.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
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Lecture Schedule:

Description	No. of Lectures
Module 1	12
Module 2	10
Module 3	10
Module 4	8